

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTECPE405B - DATA ANALYSIS

Time : 3 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which term describes the number of values that are free to vary?
a) Variance b) Confidence interval c) Degrees of freedom d) Mean
- 2 Which one is a qualitative data type?
a) Height b) Weight c) Gender d) Temperature
- 3 Missing data in a dataset can lead to:
a) Greater accuracy b) Reduced sampling error
c) Biased conclusions d) Improved inference
- 4 Which of the following is not a measure of central tendency?
a) Mean b) Median c) Mode d) Variance
- 5 Which of the following is not a measure of central tendency?
a) Mean b) Median c) Mode d) Variance
- 6 If mean = median = mode, the distribution is:
a) Skewed left b) Skewed right c) Symmetric d) Bimodal
- 7 A dataset with two peaks is called:
a) Skewed b) Symmetric c) Bimodal d) Flat
- 8 Which transformation is suitable for data with proportions or percentages?
a) Exponential b) Logit c) Box-Cox d) Freeman-Tukey
- 9 Which of the following transforms reduces variance heterogeneity?
a) Log b) Square root c) Box-Cox d) All of the above
- 10 Which test is most appropriate for testing normality in small samples?
a) Chi-square b) Shapiro-Wilk c) Z-test d) T-test
- 11 The Ryan-Joiner test is similar to:
a) Chi-square test b) Shapiro-Wilk test c) Jarque-Bera test d) KS test
- 12 Which test is used to compare two population variances?
a) Z-test b) F-test c) T-test d) Mann-Whitney test

Q2. Solve the following.

- A) Describe different types of statistical errors with suitable examples. 6
- B) Differentiate between parametric and non-parametric analysis. Give an example of when each is preferred. 6
- Q3. Solve the following.
- A) Explain the three main measures of central tendency. Discuss their merits and demerits. 6
- B) Differentiate between percentiles and quartiles. Explain how they help in understanding data spread. 6
- Q4. Solve Any Two of the following.
- A) What are power transforms? Compare them with logarithmic transforms in terms of effect on skewness. 6
- B) Describe the Box-Cox transformation. What types of data is it used for, and what is its main advantage? 6
- C) Describe the logit transformation. Where is it typically applied in data science or machine learning? 6
- Q5. Solve Any Two of the following.
- A) Compare the Anderson-Darling and Kolmogorov-Smirnov tests. In what cases is one preferred over the other? 6
- B) What is a T-test for two independent means? Describe how this test is useful when the population standard deviation is not known. 6
- C) What are contingency tables? How are they used in hypothesis testing, especially with the Chi-square test? 6
- Q6. Solve Any Two of the following.
- A) Describe the F-test for comparing two variances. What assumptions must be met for valid results? 6
- B) Explain One-way ANOVA. What are its assumptions, and how are results interpreted? 6
- C) Describe Mood's Median Test. In what situations is this non-parametric test useful? 6

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